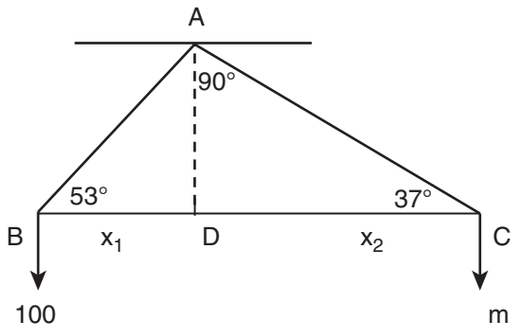
The moment of inertia of the incomplete ring is, $I = M\frac{(n-1)}{n}R^2$

7. A massless right angled triangle is suspended with its right angle corner. A mass of 100 kg is suspended from another corner B which subtends an angle 53°. Find the mass m that should be suspended from other corner C so that BC (hypotenuse) remains horizontal.

Solution:



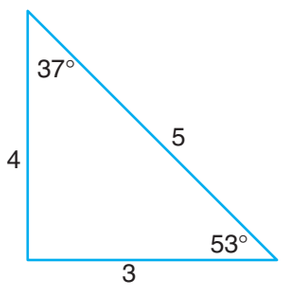
From the principle of moments,

$$100 \times g \times x_1 = m \times g \times x_2$$

$$100 \times \cos 53^\circ = m \times \cos 37^\circ \quad (1)$$

Where, x_1 and x_2 are the arm lengths.

The right angle triangle with angles 37°, 53° and 90° is a special triangle which has the respective sides in the ratio, 3:4:5 as shown in the diagram.



Substituting the values in equation (1),

$$100 \times \cos 53^\circ = m \times \cos 37^\circ$$

$$100 \times \frac{3}{5} = m \times \frac{4}{5}$$

$$m = 100 \times \frac{3}{4}$$

$$m = 75\,kg$$

8. If energy of 1000 J is spent in increasing the speed of a flywheel from 30 rpm to 720 rpm, find the moment of inertia of the wheel.

Solution:

$$\omega_1 = 30rpm = 2\pi \times \frac{30}{60} rads^{-1} = \pi \ rads$$

$$\omega_2 = 720rpm$$

$$= 2\ \pi \times \frac{720}{60} rads^{-1} = 24\pi \ rads$$

Change in kinetic energy,

$$\Delta KE = \frac{1}{2} I(\omega_2^{\ 2} - \omega_1^{\ 2})$$

$$I = \frac{2 \times \Delta KE}{(\omega_2^{\ 2} - \omega_1^{\ 2})} = \frac{2 \times 1000}{(24\pi)^2 - (\pi)^2}$$

$$I = \frac{2000}{25\pi \times 23\pi}$$

$$I \approx 0.35 \text{ kg m}^2$$

Remember:

$a^2 - b^2 = (a + b)(a - b)$

as $\pi^2 \approx 10$

9. Consider two cylinders with same radius and same mass. Let one of the cylinders be solid and another one be hollow. When subjected to same torque, which one among them gets more angular acceleration than the other?

Appendix-1.indd 287 06-01-2022 17:55:45